

Centric Unfolding

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Abstract

The question of the measurement problem in quantum mechanics has been an ongoing inquiry among physicists and philosophers for the last century or so, and has as of yet remained unsolved. In this paper, a potential solution, leaning on simple and fundamental laws of logic in our universe, will be formulated and explained. The main idea is that superposition itself is a broader concept than we have always thought, and is in fact rooted to the perceiver of their own reality more so than the cosmos itself, allowing for the same time-frozen object/concept to be in separate forms of superposition depending on the temporal iteration of the mind that is viewing it, among other things. Along the way, the concepts of knowledge, perception, and contradiction will be formulated and generalized, allowing for our basic outlook of the universe to be fundamentally charted. In the end, concepts like Schrödinger's cat and the double-slit experiment will be explained to sufficient detail regarding their existing nature.

1 Introduction

The double-slit experiment is an intriguing test which has captured the awe of physicists for over a century. When a beam of electrons is fired at a twice pierced plate, they seem to fall upon a back wall as if they had gone through both slits at once, rather than only one or the other as we might classically expect. But, perhaps eerily, when the vagueness of the unknown is challenged, and an observer of any form watches the electrons siphon through the slits, and sure enough only one slit is chosen again, causing

the pattern on the back wall to change accordingly. It may seem quite frustrating and unexplainable why this happens, but in the following pages, a pitch will be supplied for exactly why this happens, due simply to the basic power of logical reality.

2 Reality

Let us begin with an observation on the more metaphysical side of things: every comprehension we make about reality – that is every single perception – must pass through the wiring of the human brain in order for you to become aware of it. This may seem trivial at first, but if you think about it, it serves to be an extremely overbearing influence. We cannot see otherwise, and we never will, (quite by definition), and so this strict rigidity to our outlook on reality is sure to have its consequences, whatever they may be. Imagine if for every experiment you ever run, you may only observe the results through purple-colored glasses. Surely some objectivity will be lost, if there even is such a thing. There are concrete ramifications for this when regarding the human brain, such as our inability to detect neutrinos, but such concrete ramifications usually have workarounds given sufficient creativity. It is truly, then, the abstract ramifications where the real fun happens.

For the sake of this paper, I have created a simple framework to illuminate the abstract concept of knowledge. As far as I can tell, knowledge as a concept is a subclass of this logic, and not the other way around. Therefore, it must also apply to whatever it is the human brain does when generating consciousness out of observation. Imagine for a moment that a set of knowledge could be quantified. Let this set of

knowledge be called an *Ethus*. On the contrary to the sets and lists we find in computer science, an Ethus is far more organic, and yet equal in its simplicity. An Ethus is like a canteen which holds knowledge like water, and in the end it represents the collective presentation of all such knowledge. This knowledge cannot be considered to be built of bits, because such a system would fail to capture its essence¹. The Ethus contains an infinite stream of knowledge, and also a finite stream, depending simply on how one processes it. One interpretation is to consider every piece of knowledge as being part of a single, complicated property, and such a property would then be representative of the Ethus itself. We may try to describe the Ethus using language, a feat which is sometimes doable – but sometimes it is also less doable due to the abstract nature of things. You couldn't exactly describe the entire content of your memory in a sentence of two, and the same is often the case of sufficiently complex Ethuses.

If we imagine one's own mind as being a kind of Ethus, then interesting results begin to unfold. For one, the act of observing the cosmos is the act of adding to one's Ethus. In fact, our entire understanding of reality must pass at some point through our Ethus in order to be comprehended by us: and in fact it is quite possible that the two are one in the same all along. This, essentially, leads into the fundamental claim of this paper, for which the known effects of quantum physics would be a subclass: reality itself can only exist relative to an Ethus. Without an Ethus to link it to, reality is meaningless, to the point where we should not consider them separate in the first place. That is, to every Ethus, there is a reality, and to every reality, there is an Ethus – they are manifestations of each other. But trying to consider a reality without an Ethus defeats the entire point, and for that matter is physically impossible. This is an extremely counterintuitive concept to a human, so I will supply more information. Reality comes in pairs, so to speak – an observation and an Ethus working together. Crucially, reality is not a standalone thing

¹Speaking specifically, the Ethus is meant to align more with the uncountable infinity associated with probabilities more so than the countable infinity accessible by the stacking of bits.

– it must be rooted to an Ethus to have any meaning whatsoever.

3 Superposition

The next part of the story is equally interesting. Essentially, if we picture the universe as being the way it is relative to your mind's Ethus, then what happens for those inevitable properties which cannot be explained by such an Ethus? If you think about it, we can list a whole sum of properties of reality which cannot be explained one way or another using the Ethus alone, and these properties, all together, resemble an entirely new Ethus just by themselves due to their clumping nature. If we combine all of these differing properties together into a single whole, and stick it in an Ethus, then we create an *Anti-Ethus*, which is simply the partner of the Ethus. That is, for every Ethus, there exists a corresponding Anti-Ethus, which contains all the properties in reality which would have zero net effect on the Ethus if they were altered. In other words, the Anti-Ethus is statistically independent to the Ethus. As an example, if the Ethus held a property that $x = 1$, then assuming y is independent of x , the Anti-Ethus might hold a property that $y = k$, where k is a number. This property of independence still holds as the Ethus becomes more convoluted, but will of course become more convoluted itself as this happens. All this being said, for any property of reality which is partially correlated to the Ethus, but not fully, it may simply be subdivided into an Ethus component and an Anti-Ethus component which are placed into the corresponding compartments.

The key idea of the Anti-Ethus, then, is that everything within it will be in superposition when regarding reality. So, in the example of where $x = 1$ and $y = k$, it is so that literally $y = k$, in that it exists in all possible states at once. This does not, however, imply uniform distribution, because we may very well add to the Ethus that $P(y < 0) > 70\%$, for example, which would still be perfectly valid. In that case, the physical value of y is still uncertain when using the Ethus as a guide, but enough is unknown to create whatever skewed distribution would follow

given that property. In a way, the Anti-Ethus is still uniform, but only when fitting with the terrain of the Ethus. That is, the superposition of the Anti-Ethus will never arbitrarily skew one way or another without the direct order of the Ethus to do so. If the Ethus is a container, then the Anti-Ethus fits that container like a liquid.

4 Contradiction

And that leaves one final important rule of reality: *no contradictions are permitted*. Even if a contradiction occurs somewhere in the infinite superposition of an Anti-Ethus upon reality, then something is wrong, and so reality will take the simplest possible path to avoid that, adding to the Ethus accordingly. Note that a contradiction may be defined as an Attribute in the Ethus holding two possible values instead of just one – which unlike the Anti-Ethus, is strictly forbidden. This includes a scenario where a property in the Anti-Ethus conflicts with its corresponding occurrence in the Ethus. Either way, the Ethus in question is invalid². If we picture a child Ethus as meaning an Ethus with added information from another Ethus, then a parent Ethus would be the product of removing information from an Ethus. Contradictions cannot trigger on the usual plane of reality, where an Ethus is center stage, but technically such a concept is only a special case of a more general concept: if an Ethus with removed information from another causes a contradiction, then both Ethuses are invalid. Another way of putting it is that every child of an invalid Ethus is also invalid³.

²Even though an Anti-Ethus, in its definition, is assumed to be statistically independent of the Ethus, it still holds that any Anti-Ethus which clashes somewhere with its Ethus, hypothetical or practical, would be collectively invalid in the real sense.

³Note that any perceived logical deviation between this and the previous postulates is only an illusion, and in actuality they play off of each other quite nicely. If anything, a contradiction to an Ethus in its Anti-Ethus is proof that such an Anti-Ethus is not applicable to the Ethus in the first place, defeating its definition of independence. This is another way of looking at invalidity. Then, from there, the argument that the child of any invalid Ethus is also invalid highlights the fact that covering up a contradiction by setting an Ethus property

5 Assembly

Putting all this together allows for three fundamental rules of reality.

1. Anything may only be real relative to an Ethus.
2. Any attribute of an Anti-Ethus will be in superposition relative to its Ethus.
3. An Ethus is invalid if a contradiction is triggered anywhere in the superposition of its parent Ethuses.

These three basic laws of Ethus manifestation are sufficient to describe all superposition related queries, and furthermore make some intriguing implied claims. For one, there is no objective reality of the universe, as if to say that a particle in superposition is such relative to the universe. Instead of this, any superposition only relates to a corresponding Ethus, which, crucially, may very well differ between people. If an experimenter is unaware of the state of Schrödinger's cat, then relative to their Ethus, the cat is in superposition of being alive and dead, because such a state sits in the Anti-Ethus of the experimenter's mind. However, even if such an experimenter looks in the box and becomes aware of the answer, the cat will still be in superposition relative to an unknowing bystander, and as a matter of fact, so too will the experimenter, because that bystander does not know how they reacted. However, the second the experimenter tells the bystander how the experiment went, and they will also add the information to their Ethus – however not entirely. Technically there is a chance the experimenter is lying, and for that matter, there is a chance the cat quantum fluctuated to an alternate state in the time it took for the bystander to become informed. Therefore, it is physically impossible to perfectly add a state to the Ethus at some certainty one way or another.

to a set value does not fix the fact that the property is still wired to potentially trigger a contradiction in the first place. Thus, an invalid Ethus, in being functionally contradictory, may only beget invalid children Ethuses. Therefore, to check an Ethus for validity, one must prove that every possible parent/ancestral Ethus to it will never trigger a contradiction to its Anti-Ethus.

A superposition may never truly collapse – rather it can only compress⁴. All of this being said, the precise breakdown of Schrödinger’s cat according to the Ethus logic is that A) The cat was only in superposition⁵ relative to the experimenter’s unknowing Ethus before they checked the box, (and crucially, the cat remained in superposition forever beyond then relative to such a past Ethus), and that B) The cat was in the same state the experimenter observed the entire time, relative to their Ethus at the temporal moment of and following the observation. This logic allows us to generalize superposition and also perhaps make it more compatible with Einsteinian thought in the process, because information perhaps cannot travel between two realities at all, not to mention faster than light. The possibilities here are intriguing, but for the remainder of this paper we will continue to hug the propositions of these three rules alone, and see where it leads.

6 Thought Experiments

Take, for another example, the double-slit experiment itself. Let us imagine running a dual thought experiment here: one where the particles are electrons in a laboratory, and another where they are humans on a soccer field. Set up on this field is every needed analogue to the normal experiment, except instead of two slits on a plate, there are two doors on a long constructed wall, which itself is in front of another wall with numbered labels from one to a hundred along it. In front of the two doors, concealed from the back wall, are one hundred participants eager to mimic electrons, and standing blindfolded on the bleachers is our classic quantum observer. Let the games begin!

Every participant in the soccer field holds a smartphone with two numbers texted to them: the first is their door to travel through – either one or two – and the second is the spot on the back wall where they are to go stand. Both numbers are calculated precisely randomly given the setting, just as would be the case for the normal double-slit experiment, which

means that if the people did indeed go through both doors at once, they would also appear in an identical wavelike pattern. Each person awaits their text, and then upon receiving, heads through the corresponding door(s) and stands at the back wall. This goes on for twenty minutes, and soon the experiment has been run. Now all that is left is for the blindfolded quantum observer to see the results. They walk, still blindfolded, to the back wall, and then on the count of three, remove the blindfold to see... two major regions of people on the back wall, which does not signify superposition at all. So, what went wrong?

The red herring here is to assume the people never were in superposition. Relative to the observer right now who looks at the wall, no, the people are not in superposition, however relative to the observer ten minutes ago when they were blindfolded over by the bleachers, the people were indeed in superposition. Technically those are two separate Ethuses, and so they may very potentially correspond to separate realities⁶.

⁶In other words, the superposition did not update with time by some objective occurrence – rather only the illusion of this triggered due to the update of the observer’s Ethus. Even now, as the experiment has finished, relative to the observer’s Ethus ten minutes ago, all these people are in superposition. And when looking backwards, the observer’s current Ethus knows that each person went through one and only one door to satisfy rule number three, however they are still unsure of which door it was until they ask someone or otherwise reference some other clue. To be clear, the act of a participant ten minutes ago going through both doors at once relative to the observer in the present would break rule number three, so, assuming the validity of this paper, this would make such a concept invalid. At the same time, however, the observer may very well not know which exact door the participant went through then, even if a good amount of information is deductively available given the current location of said participant. Thus, the observer’s Ethus knows that the participant went through one door ten minutes ago, but they also are mentally unsure of which door it may have been to a precise degree. This seems like a kind of paradox, because supposing that the observer is unsure of the door the participant took seems to imply that such information would be in superposition by rule number two. Technically speaking, however, there are a list of subtle workarounds to this paradox. For one, perhaps the observer cannot be sure the participant went through only one door, due to the outside chance that the universe corrected for itself in the moment by some puzzling quantum phenomenon, which disallows the third rule to be broken by concealing the exact right information in sequence. This cannot be ruled out as a possibility because of

⁴This will be referred to as the *compression postulate*.

⁵Other than the compression postulate’s interference of course.

When the observer was blindfolded ten minutes ago, and the experiment was still running, each new participant who walked through a door did so without the knowing of the observer, and so as far as their Ethus could tell, they were walking through both doors at once. Only when the observer bothered to look at the results did the superposition compress into the result. The observer at that moment could not see the numbers on the participants' screens, so still the actual door index of each person is unknown, but nonetheless the fact that no single person traveled through two doors is certain to them now that they see the pattern of where they stand. The reason for this is quite simple: if the observer's Ethus was aware that each person went through both doors, due perhaps to a wavelike pattern on the back wall, then if they were to ask each person which door they went through, surely gathering one door for each, the results would be contradictory. The observer may not currently be speaking to another person to gather information, but if we create a parent Ethus where this known is removed, contradiction will instantly trigger somewhere in the superposition if there is a wavelike pattern of people on the back wall. Thus, rule number three would be broken if such an Ethus was permitted, and so, blatantly speaking, it is not. Therefore the observer physically cannot observe a wavelike pattern of people on the back wall, (without some kind of weird fluke obviously).

But on the microscopic scale, of course, is where everything gets weird. One may be inclined to call the quantum world an entirely separate system abiding by entirely separate rules, but by the logic of this paper, that would be overthinking it. Truly, then, the same exact principles are working down there as they

are up here, but with one red herring-typed catch: Heisenberg's uncertainty principle. The idea is actually stupidly simple when you look at it closely – so long as the electrons were not observed when going through the slits, then by the uncertainty principle, we may determine that that information is permanently lost. Therefore, we may remove attributes at will from our Ethus as the observer, and never once will a contradiction occur. Therefore, calling the electrons in a superposition long after the experiment is over is totally valid, and so that they will be just that. If, however, we observe the electrons going through the slits, then when we observe the pattern later on, a wavelike result would break rule number three. We obviously cannot break rule number three, and so such an Ethus cannot take the form of one's mind. Instead, we will have to observe the particles in the form of the two groups, just as we did with the people. As for the exact reason why such a thing would break rule number three, it is because if we imagine a parent Ethus to the observer's where the knowledge of whether or not they checked the detector is removed, then in certain regions of the superposition of that they have seen the electrons go through one slit or the other⁷ in addition to seeing the wavelike pattern on the wall⁸. Of course this is a contradiction, and so as long as the detector was active during the experiment, the particles must form two groups on the back wall. The idea, though, is that all one has to do is feed Heisenberg's uncertainty principle into this Ethus logic, and the precise conclusions of the double-slit experiment will be generated. And the added bonus is that we could also just not, specifically when regarding the macro scale, and then those results will follow as well.

the compression postulate. But there are other fixes to this paradox as well, such as that the Ethus conceals information from a mind in just the right way to counter such an example – perhaps in some eerie way, every parent Ethus to your own is equally real in some sense, and so what you physically observe of the universe is covered by whatever Ethuses lie above you in the “tree”. Obviously this is quite speculative, however, and strange for it. Both of these fixes feel somewhat unsatisfactory however for their lack of simplistic clarity, so the real answer could very well be distinct and more clever than either of them, such as that some placeholder data is present in the Ethus at the time, or something else entirely. Let the readers be the judge of that.

7 Human Experience

Another quick topic of interest might be how this all relates to the human experience. You would be crazy, as a human, not to consider everything you have read some form of absurd. But then again, it probably is not absurd in actuality. We are specifically pro-

⁷By virtue of checking the detector's results.

⁸Implying that the electrons went through both slits.

grammed by paleomammalian environmental evolution to think it is anyway, though, because if we were inherently open to this Ethus-oriented perception of the world, we would have been crushed by the cruel grasp of mental evolution ages ago. So in a way every one of us is living a lie every time we assume that reality is A) objective, and B) not consistently riddled with superpositions. I am not recommending that you saturate your social life with these new truths, because frankly that will probably lead any one of us to madness, but still it's not all bad. This Ethus logic is in no way shape or form in favor of solipsism, for a few key reasons. For one, every person you interact with has their own Ethus, every one of which is equal in its dignity to your own. The thing is, you are simply seeing a reality relative to *your* Ethus, however all of them are seeing a reality relative to their Ethuses as well. Interestingly, the various superpositions in these realities will never quite line up, but both are equally valid with respect to each Ethus – you just happen to be in yours and they happen to be in theirs. It is like the multiverse all played out right in our own world. And no, this is not anthropocentrically based. Any old knowledge collecting entity in the universe may have an Ethus if you suppose, and for that matter even abstract things like thought experiments may. It's just that every time you're not aware⁹ of something, then when wiring to your own Ethus, that thing is then in superposition. But relative

⁹As for how this relates to the mind itself is still uncertain, specifically regarding the distinction between thinking one sees a bear, and seeing a bear, for instance, to give an evolutionarily relevant take. The Ethus, in this case, would be the fact as opposed to the (conscious) perception of the fact. Nonetheless, it seems, this would be a most organic fact, because it is strictly correlated to the mind itself in the case of something like a human. The only input to the Ethus is the capabilities of one's senses themselves, and by the first rule of the Ethus, such a thing shapes reality around the observer. Not nearly enough is known about the mind at this point to make a definitive claim, but perhaps the Ethus serves as a kind of foundation for the subconscious mind, like an iteration upon iteration lower than what is consciously available to a human at any given moment, itself serving as a big part of our reasoning. Or, maybe the answer is something else entirely, but regardless of what we find, the Ethus is only ever the ultimate and most fundamental iteration of a human mind's knowledge, even if that means it doesn't always align with what you think you know.

to any example of what that thing is, you yourself may be in superposition. And remember that key time component as well, which states that the same mind at different times is a different Ethus. At the moment of your roommate going to work while you stay home, they may be in superposition regarding certain attributes, but once they come home and tell you about their day, now relative to your new Ethus, they were not in superposition during that workday.

Also regarding the human experience is our imaginations. Our capacity to imagine is designed by evolution to fit with our funny mammalian outlook on the universe where everything is objective. Back in the soccer field experiment, when the blindfolded observer is sitting patiently on the bleachers during the experiment, they may try to imagine what is going on over by the doors. They may imagine a person going through each door, and they imagine the people settling over at the walls. Perhaps they ask the question, "What's happening over there right now? What does the pattern look like?" But, now, with this Ethus logic, the answer is strange – essentially there is no pattern, because even imagining one is adding more knowledge to their Ethus. In a sense, the person has generated an entire pretend Ethus to try and grasp what is going on, and a pretend Ethus which has no guaranteed logical validity at that. In actuality, the events of the doors and the back wall at the time, while in superposition, cannot be observed in the first place while the observer is in their current blindfolded state. So the answer to what it looks like is that it looks like nothing, because such a human-bias-oriented question defeats the purpose. Even more technically speaking, the act of "looking" is the act of bouncing photons off an object and catching them in one's eyes. But, well, such a concept is immaterial and irrelevant when regarding a superposition. This does not mean that the people over there are mindless ghosts or something, but rather they are doing every possible thing they could be doing at once, all depending on the various likelihoods predefined in the observer's Ethus.

8 Conclusion

So, the moral of the story is that all of this Ethus logic is just quite strange, and that we ourselves are strangely resistant to it by nature. We may one day know the great truths of physics and metaphysics, but in the meantime, all we can do is keep asking questions, and crucially, keep clearing away the clouds of bias which have locked up our perception for millennia. Maybe, then, we will finally see the sun which illuminates our every passing second in this extraordinary universe.